

2016.0 RANGE ROVER (LG), 501-30

REAR END SHEET METAL REPAIRS

QUARTER PANEL (G1454642)

REMOVAL AND INSTALLATION

REMOVAL

NOTES:

- The quarter panel is installed in conjunction with:
 - Rear bumper
 - Loadspace trim panel
 - C-pillar lower trim panel
 - D-pillar trim panel
 - Fuel tank filler pipe
 - Rear quarter window glass
 - Rear safety belt retractor
 - **Right hand indicated, left hand similar.**
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1.



E143429

The quarter panel is serviced as indicated.

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2. Before commencing this procedure make sure that you are aware of all Health and Safety requirements. For additional information, refer to: [Health and Safety Precautions](#) (100-00 General Information, Description and Operation).
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3. For further information on the methods, tools and fixings used in this procedure refer to the body repairs - general information section.

NOTES:

- Where breakstem fasteners (BSF) removal is instructed remove fastener shaft with a 4mm punch and remove the head with a 6.5mm drill bit. Make sure all the drilled holes are free from debris.
- Where self piercing rivets (SPR) removal is instructed use either the approved SPR tool or 5.3mm drill bit.
- Any joints sealed as part of vehicle manufacture must be sealed as part of the repair.

4. For additional information, refer to: [Rear Bumper](#) (501-19 Bumpers, Removal and Installation).

5. For additional information, refer to: [Loadspace Right Trim Panel](#) (501-05 Interior Trim and Ornamentation, Removal and Installation).

6. For additional information, refer to: ['C' Pillar Lower Trim Panel](#) (501-05 Interior Trim and Ornamentation, Removal and Installation).

7. For additional information, refer to: ['D' Pillar Trim Panel](#) (501-05 Interior Trim and Ornamentation, Removal and Installation).

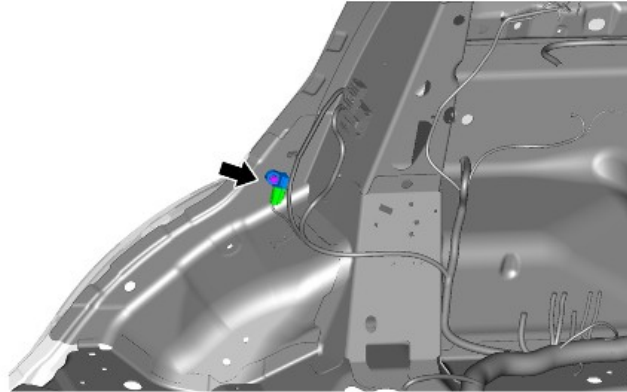
8. For additional information, refer to: [Rear Quarter Window Glass](#) (501-11 Glass, Frames and Mechanisms, Removal and Installation).

9. For additional information, refer to: [Rear Safety Belt Retractor](#) (501-20A Seatbelt System, Removal and Installation).

10. For additional information, refer to: [Fuel Tank Filler Pipe](#) (310-01A Fuel Tank and Lines - TDV6 3.0L Diesel, Removal and Installation).

11. Release the load space wiring harness and position to one side.

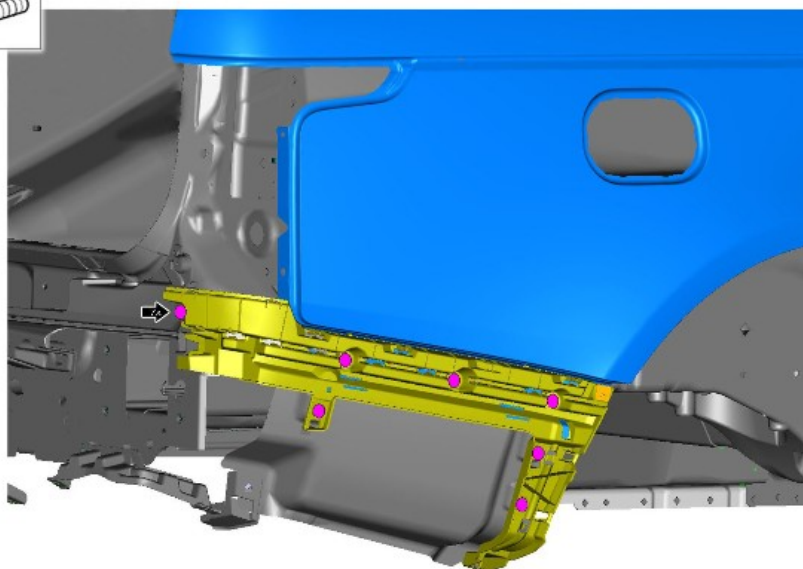
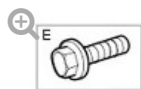
12.



E144388

Remove the C-pillar side impact sensor as indicated.

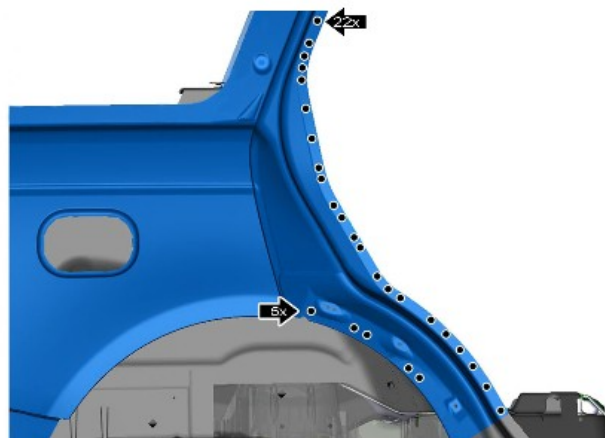
13.



E144395

Remove the rear bumper cover side retaining bracket as indicated.

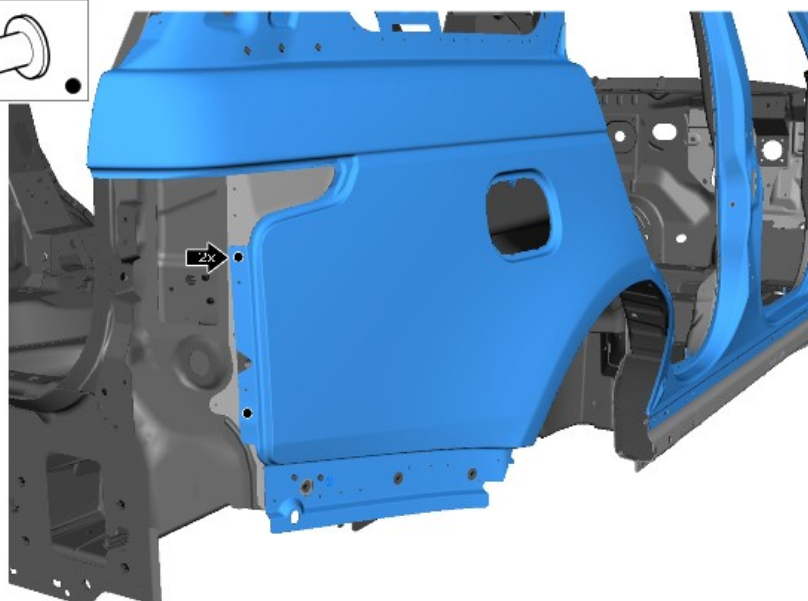
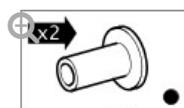
14.



E143432

Remove the SPR's as indicated.

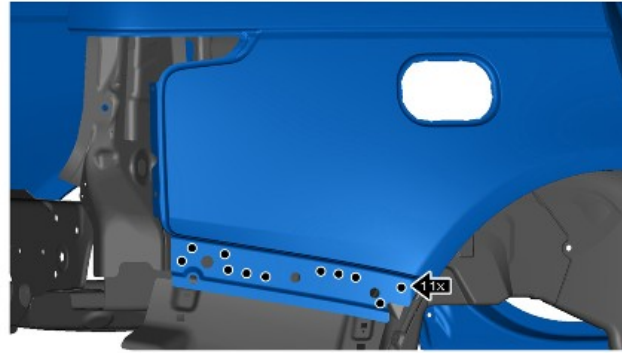
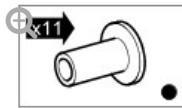
15.



E143431

Remove the SPR's as indicated.

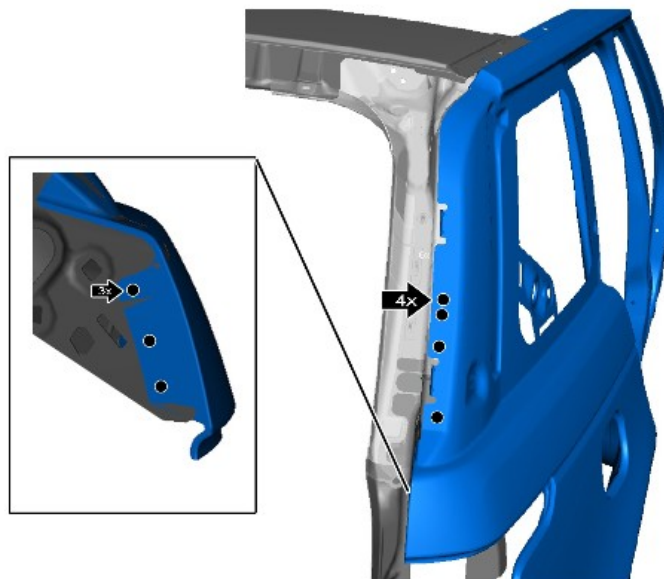
16.



E143435

Remove the SPR's as indicated.

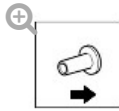
17.



E143434

Remove the SPR's as indicated.

18.



E143438

Remove the SPR's as indicated.

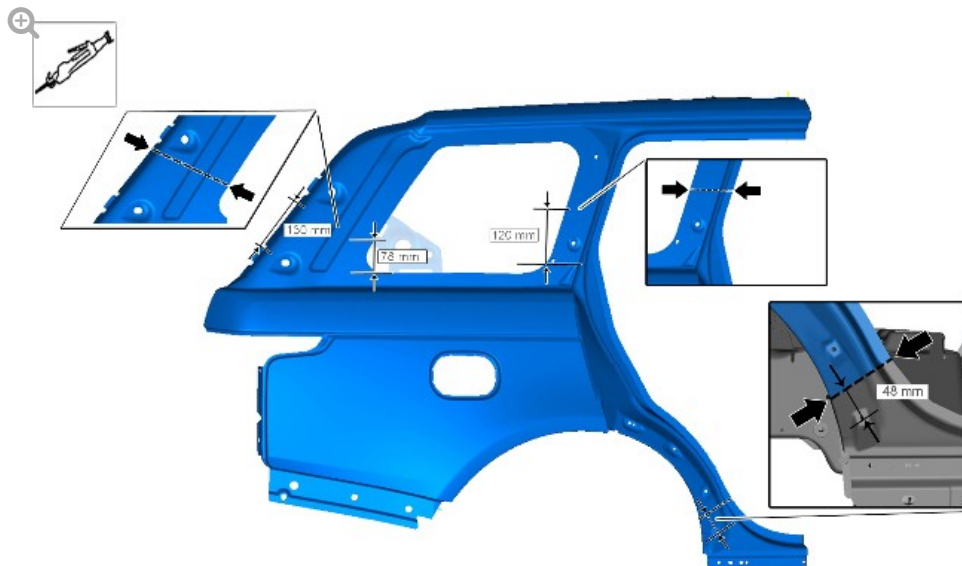
19.

CAUTION:

Where possible avoid damaging the inner panels when cutting the quarter panel.

NOTE:

The cut lines shown are the maximum range allowed.



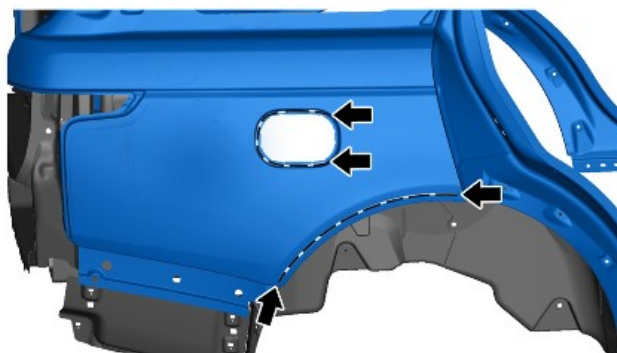
E143430

Mark and cut the quarter panel as indicated.

20.

NOTE:

Noise, vibration and harshness (NVH) components are located behind the quarter panel.



E143436

Carefully separate and release the adhesive between the quarter panel and the rear wheelhouse outer, rear outer lamp panel outer

and the fuel filler flap aperture as indicated.

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21. Separate the joints and remove the quarter panel and release the NVH components.
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INSTALLATION

CAUTION:

Where structural adhesive is to be applied between surfaces it is essential compression is applied to the joints during the curing period.

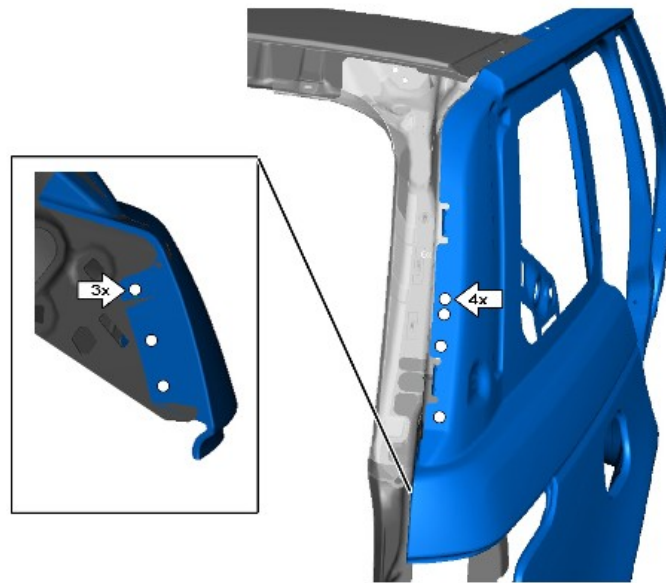
NOTES:

- BSF installation involves using the Land Rover approved BSF installation tool and requires a hole diameter of 6.5mm.
- SPR installation requires using the Jaguar approved SPR tool.

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1. Offer up the new quarter panel and mark the cut lines on the new quarter panel.
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2. Trim the excess from the new quarter panel.
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3. Remove rivet remnants.
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4. Using a fine bristle disc, remove the adhesive residue.
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5. Dress flanges where necessary.
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6. Offer up the new quarter panel and clamp into position. Check

alignment, if correct proceed to next step, if not, rectify and recheck before proceeding.

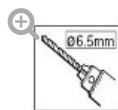
7.



E 144397

Drill holes through the old quarter panel into the new quarter where BSF's are to be installed as indicated.

8.



E 144530

Drill holes through the old quarter panel into the new quarter where BSF's are to be installed as indicated.

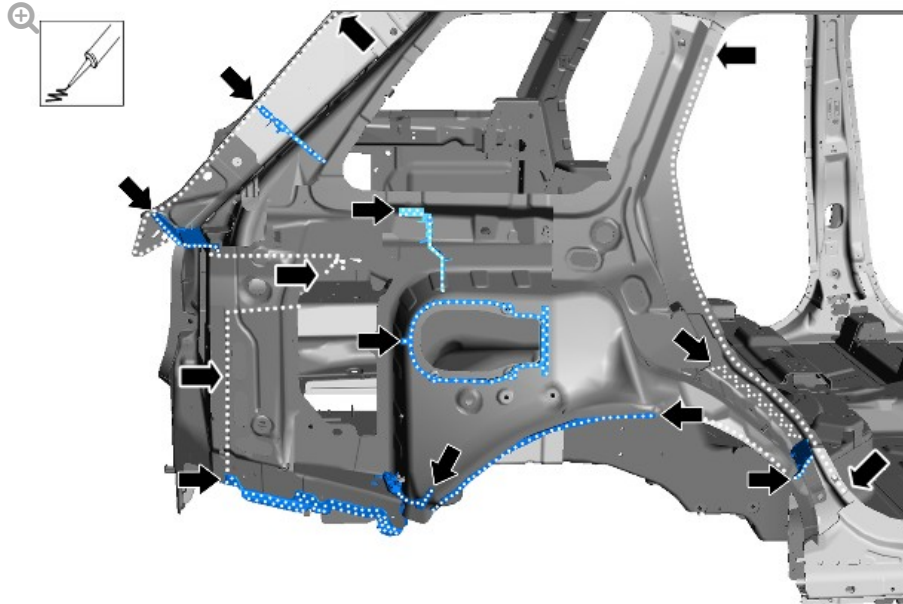
9. Remove the new panel.

10. Deburr the drilled holes.

11. Using a fine bristle disc, clean and prepare the panel surfaces.

12. Apply the coupling agent where the Land Rover recommended bonding material is to be applied and allow to dry.

13.



E143437

Apply a bead of Land Rover recommended semi-rigid sealer to the rear wheelhouse outer and the fuel tank filler pipe housing as indicated.

14.

CAUTION:

Make sure that adhesive is not applied within 30mm of the metal inert gas (MIG) welded butt joints.

NOTE:

Make sure a continuous bead of adhesive surrounds the fixing holes.

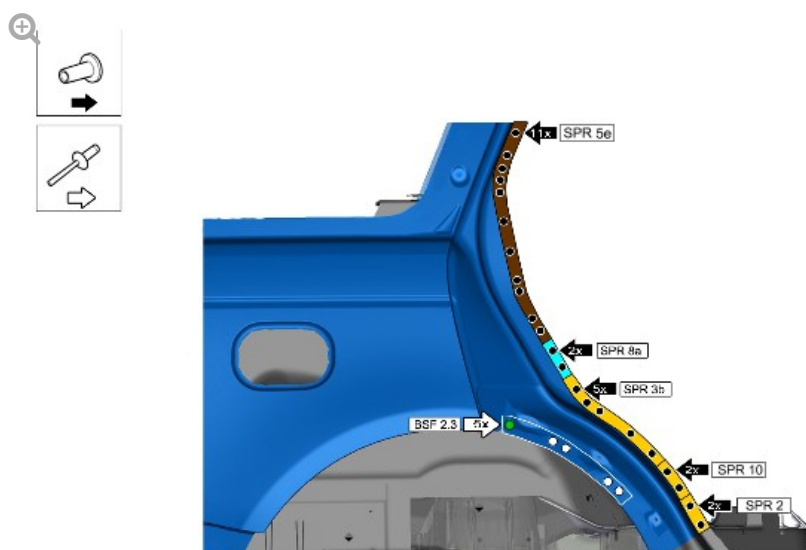
Apply a 5mm zig zag bead of the Land Rover recommended bonding material to the body joints.

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15. Offer up the new quarter panel and clamp into position. Check alignment, if correct, proceed to next step, if not rectify and recheck before proceeding.
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16.

NOTE:

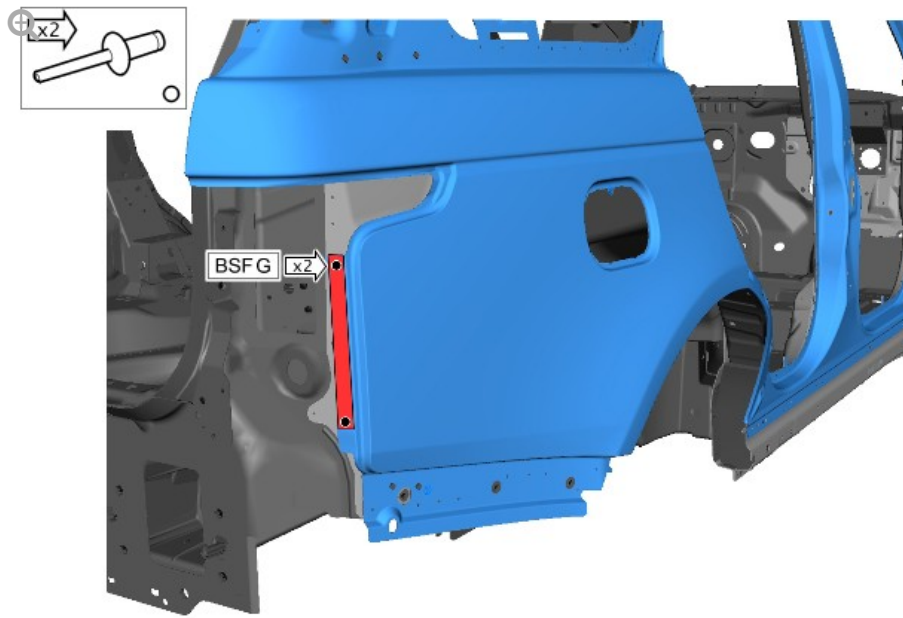
Make sure that the BSF highlighted in green is installed from the inside.



E144393

Install the SPR's and the BSF's as indicated.

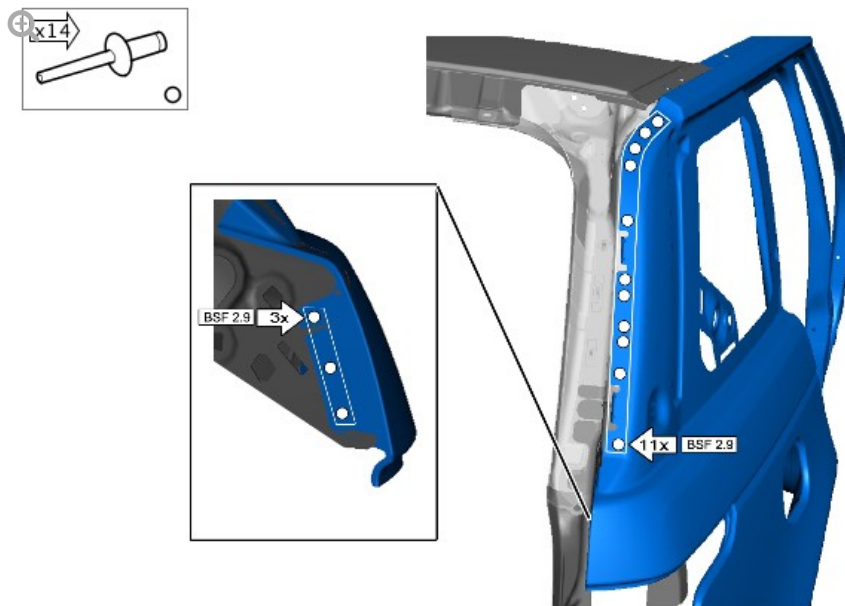
17.



E144389

Install the BSF's as indicated.

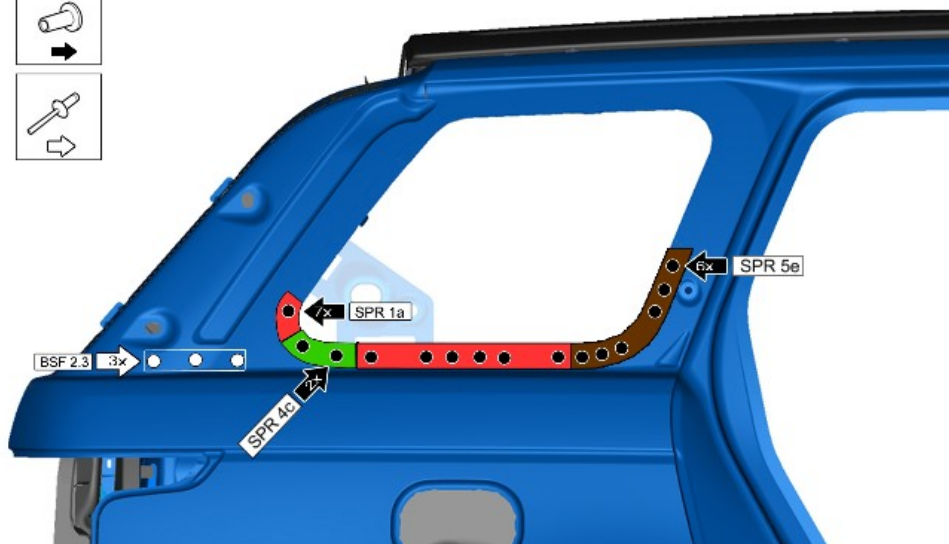
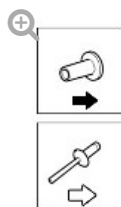
18.



E144391

Install the BSF's as indicated.

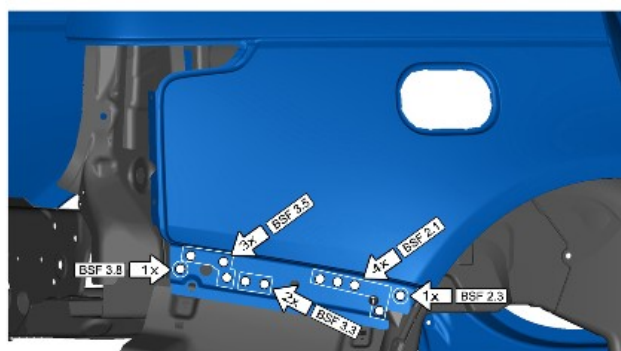
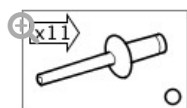
19.



E143433

Install the SPR's and the BSF's as indicated.

20.

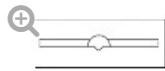


E144394

Install the BSF's as indicated.

21. Remove any excess adhesive.

22.



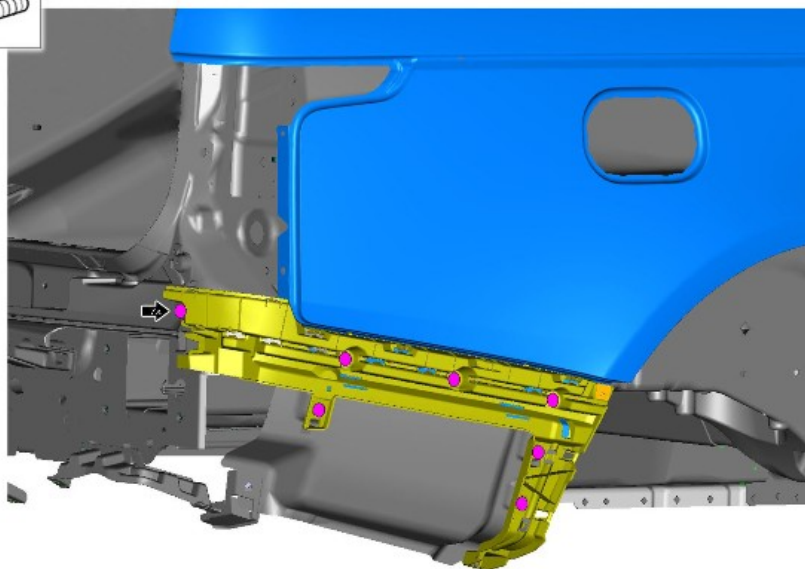
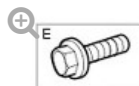
E144392

MIG weld the quarter panel butt joints as indicated.

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23. Carry out a non destructive crack inspection on the MIG welded butt joints. If correct proceed to next step, if not, rectify and recheck before proceeding.
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24. Dress the welded joints.
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25.



E144395

Install the rear bumper side retaining bracket and tighten the bolts

to 10Nm.

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26. Make sure that any open or exposed panel joints are correctly sealed.
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27. Make sure corrosion protection is applied to all areas affected by repair.
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28. The installation of associated panels and components is the reversal of removal procedure.